Type MS · Convert between pie-chart sectors, percentages and counts

Answer 1 of 4 · Question 8 part b.i · 3 marks

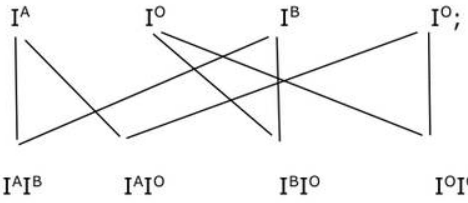
2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally; • artificial passive immunity are when antibodies are injected into the body;	accept through breast milk/placenta accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 7 · find part a.i in this mark scheme extract

4 (a)	(i)	**Africa** Blood group / Percentage of blood group O: 68 A: 17/17.5 B: 10; AB: 5 **Europe** Blood group / Percentage of blood group O: 45 A: 40 B: 12 AB: 3;		2
	(ii)	• reference to antigens on red blood cells (group B); • antibodies produced; • causing agglutination/clumping (of red blood cells);		3
	(iii)	I^AI^O I^BI^O;  I^A I^O I^B I^O; I^AI^B I^AI^O I^BI^O I^OI^O;	Full marks for a Punnett square if parent genotypes are given.	3
	(iv)	1 in 4:/25%/¼/0.25;		1
(b)		• <u>deoxygenated</u> blood moves into the left side/ventricle of the heart; • (deoxygenated blood) mixes with <u>oxygenated</u> blood; • less oxygen travelling around the body/to body/cells/in bloodstream; • breathes faster to get more oxygen into the body;		4
				Total 13

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 7 · find part a.i in this mark scheme extract

	Answer	Notes	Marks

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 3 of 7 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks
6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks
7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 4 of 7 · find part a.i in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 5 of 7 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 6 of 7 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks
9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 2 of 4 · Question 4 part a.i · 2 marks

2022 JUN · Paper 01 · extract 7 of 7 · find part a.i in this mark scheme extract

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Type MS · Convert between pie-chart sectors, percentages and counts

Answer 3 of 4 · Question 6 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 4 of 4 · Question 4 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Question number				
4	(a)	• correct plots;;; • key/labels;		4
	(b)	• correct components/nutrients/list of components min of five; • in correct amounts/proportions;		2
	(c)	• 75; • 15 • = 5:1;	ECF	2
	(d)	• contain protein • needed for growth/repair/,muscles/tissues; • area A girl has low(er) levels/area B girl has high(er) levels; • less/poor growth/area B girl good growth; • kwashiorkor;		5

Total 13 marks

Type MS · Convert between pie-chart sectors, percentages and counts

Answer 4 of 4 · Question 4 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks